

ELECTRIC POTENTIAL

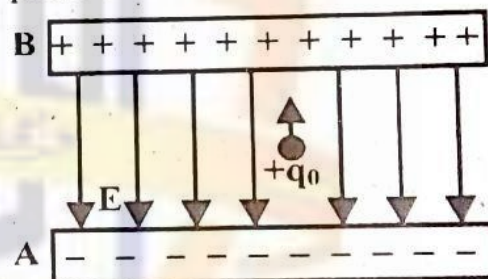
Q.1 What is electric potential energy ?

ELECTRIC POTENTIAL ENERGY

Consider a $+q_0$ test charge is placed between two oppositely charged plates A and B. The electric field lines come out from positive plate and enter into negative plate.

The test charge $+q_0$ will move along electric field lines and gains kinetic energy because positive plate B repels charge $+q_0$ while negative plate A attracts it.

The work done W by an external force is needed to move test charge $+q_0$ against electric field between initial point i and final point f which lies on plate A and plate B respectively. This negative work done is equal to difference in electric potential energy between two points.



$$\Delta U = U_f - U_i$$

$$\Delta U = -W$$

Where U_i and U_f are potential energies at initial position i and final position f respectively. The potential energy of a charged particle in an electric field depends upon the magnitude of charge. If we set $U_i = 0$ at infinity, then potential energy at final position f is

$$U_f - 0 = -W$$

$$U_f = -W$$

Q.2 What is electric potential energy in terms of electric field? Calculate electric potential energy of a system of charged particles.

ELECTRIC POTENTIAL ENERGY

The electric potential energy of a system of fixed point charges is equal to the amount of work done needed by an external agent to assemble the system.

It is necessary that each charge is brought from infinite distance and the charges are at rest in their initial positions and in their final positions.

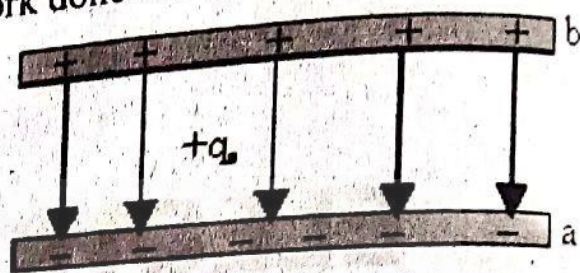
Consider a $+q_0$ test charge is placed between two oppositely charged plates. The difference in potential energy ΔU when q_0 moves between point "a" and "b" under the action of electric force $F = q_0 E$ is equal to negative work done and given as

$$\Delta U = -W_{ab}$$

This equation can be written as

$$U_b - U_a = - \int_a^b \vec{F}_e \cdot \vec{ds}$$

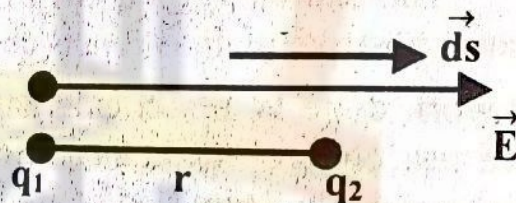
$$U_b - U_a = - \int_a^b q_0 \vec{E} \cdot \vec{ds}$$



When we have infinite separation between plates, the potential energy is zero at some reference point, so electric potential energy is written as

$$U = - \int q_0 \vec{E} \cdot \vec{ds}$$

Consider two charges q_1 and q_2 separated by distance r as shown in fig. The component of electric field along line is



$$E = \frac{k q_1}{r^2}$$

The change in potential energy is

$$U_b - U_a = - \int_a^b q_2 E dr$$

Where charge q_2 moves relative to $+q_1$ through displacement ds .

$$U_b - U_a = - \int_a^b \frac{k q_1 q_2}{r^2} dr$$

$$U_b - U_a = -k q_1 q_2 \int_a^b r^{-2} dr$$

$$U_b - U_a = -k q_1 q_2 \left[\frac{r^{-2+1}}{-2+1} \right]_a^b$$

$$U_b - U_a = k q_1 q_2 \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

If we choose reference point 'a' at infinity,

$$U_b - U_\infty = k q_1 q_2 \left[\frac{1}{r_b} - \frac{1}{r_\infty} \right]$$

$$U_b - 0 = k q_1 q_2 \left[\frac{1}{r_b} - 0 \right]$$

$$U_b = \frac{k q_1 q_2}{r_b}$$

The general formula of electric potential energy is

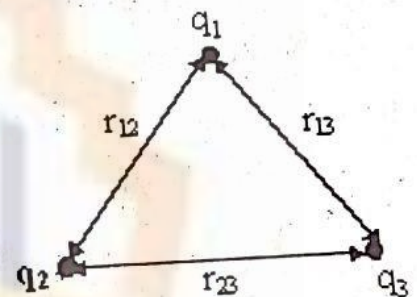
$$U(r) = \frac{k q_1 q_2}{r}$$

The electric potential energy is negative if the electric force is attractive. It means sign of q_1 and q_2 is opposite. The electric potential energy is positive if the electric force is repulsive.

Consider three charges q_1 , q_2 and q_3 having separation r_{12} , r_{13} and r_{23} as shown in fig. The electric potential energy of the system is

$$U(r) = U_1 + U_2 + U_3$$

$$U(r) = \frac{k q_1 q_2}{r_{12}} + \frac{k q_1 q_3}{r_{13}} + \frac{k q_2 q_3}{r_{23}}$$



Q.3 What is electric potential?

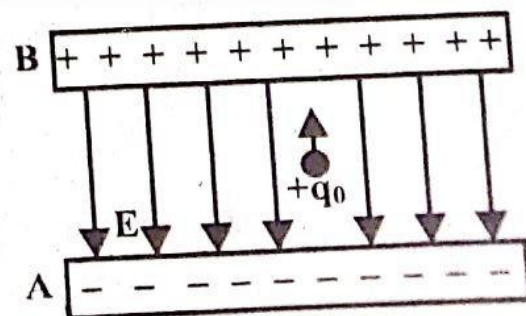
ELECTRIC POTENTIAL

Consider a $+q_0$ test charge is placed between two oppositely charged plates A and B. The electric field lines come out from positive plate and enter into negative plate. The work done W by an external force is needed to move test charge $+q_0$ against electric field between initial point i and final point f which lies on plate A and plate B respectively. This negative work done is equal to difference in electric potential energy between two points.

$$\Delta U = U_f - U_i$$

$$\Delta U = -W$$

The negative work done by the electric force to move unit test charge with constant velocity from one point to other point is called potential difference between these points. The potential difference can be positive negative or zero depending on the sign and magnitude of charge and work.



$$\Delta V = -\frac{W}{q_0}$$

$$V_f - V_i = -\frac{W}{q_0}$$

The electric potential V_f is zero when final position f is shifted at infinity. Hence electric potential at any point in an electric field is

$$0 - V_i = -\frac{W}{q_0}$$

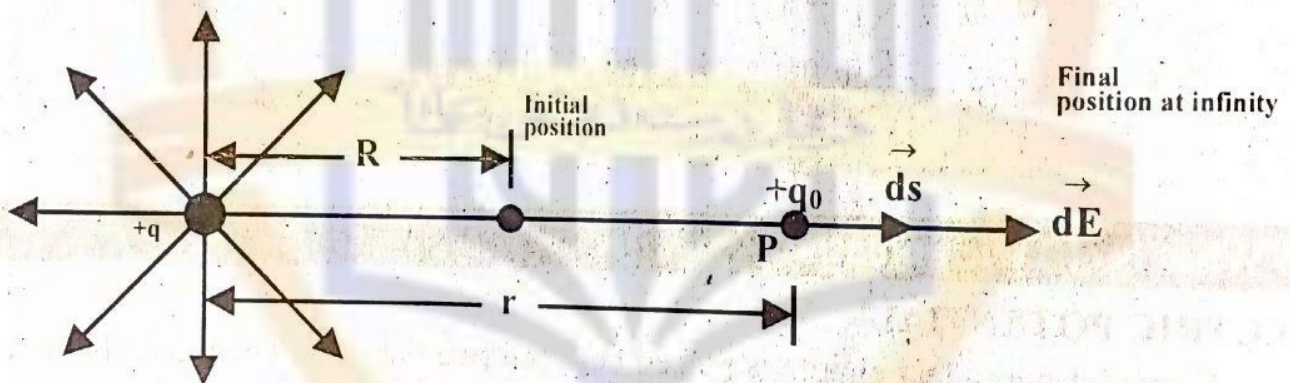
$$V = \frac{W}{q_0}$$

Q.4 Calculate electric potential due to point charge and N-point charges.

ELECTRIC POTENTIAL DUE TO POINT CHARGE

The electric field lines come out radially from positive charge. Consider such a line which extends radially from charge $+q$ through point P to infinity distance.

Now place a test charge $+q_0$ at point P having distance r from charge $+q$. The charge $+q$ repels test charge $+q_0$ with electric force $\vec{F}_e = q_0 \vec{E}$.



The work done on test charge $+q_0$ due to electric force $\vec{F}_e$ when it moves $+q_0$ through displacement $\vec{ds}$ between initial position and final position is

$$W = \int_i^f \vec{F}_e \cdot \vec{ds} = q_0 \int_i^f \vec{E} \cdot \vec{ds}$$

$$\frac{W}{q_0} = \int_i^f \vec{E} \cdot \vec{ds}$$

(1)

The negative work done by the electric force to move a unit charge with constant velocity from one point to other point in electric field is called potential difference between these two points.

$$\Delta V = -\frac{W}{q_0}$$

$$\frac{W}{q_0} = -(V_f - V_i) \quad \text{----- (2)}$$

Comparing eq(1) and eq(2)

$$V_f - V_i = - \int_i^f \vec{E} \cdot \vec{ds}$$

$$V_f - V_i = - \int_i^f E \, ds \cos \theta$$

The angle $\theta = 0^\circ$ because electric field $\vec{E}$ and displacement $\vec{ds}$ have same direction. Since path is radial so put $ds = dr$

$$V_f - V_i = - \int_i^f E \, dr \cos 0^\circ$$

$$V_f - V_i = - \int_i^f E \, dr$$

The magnitude of electric field at the site of test charge $+q_0$ is $E = \frac{kq}{r^2}$

$$V_f - V_i = - \int_i^f \frac{kq}{r^2} \, dr$$

Now put $V_f = 0$ at distance infinity and $V_i = V$ at distance R .

$$0 - V = - \int_R^\infty \frac{kq}{r^2} \, dr$$

$$V = \int_R^\infty kq r^{-2} \, dr$$

$$V = kq \left| \frac{r^{-2+1}}{-2+1} \right|_R^\infty = -kq \left| \frac{1}{r} \right|_R^\infty$$

$$V = -kq \left(\frac{1}{\infty} - \frac{1}{R} \right)$$

$$V = -kq \left(0 - \frac{1}{R} \right) = \frac{kq}{R}$$

This electric potential $V = \frac{kq}{R}$ is at initial position. In general, electric potential at any point P having distance r from positive charge q is given as

$$V = \frac{kq}{r}$$

Similarly, electric potential at any point P having distance r from negative charge q is given as

$$V = \frac{-kq}{r}$$

It means positive charges produces positive electric potential and negative charge produces negative electric potential.

UNIT

The SI unit of electric potential is called volt and written as

$$\text{volt} = \frac{\text{joule}}{\text{coulomb}}$$

Generally, the word voltage is used for electric potential or potential difference. The formula for change in potential energy is

$$\Delta U = q_0 \Delta V$$

The unit of electric potential energy is joule when voltage is measured in volts and charge is measured in coulombs. The unit of electric potential energy is generally electron-volt (eV) when voltage is measured in volts and charge is measured in terms of elementary charge e .

ELECTRIC POTENTIAL DUE TO COLLECTION OF N-POINT CHARGES

Consider charges $q_1, q_2, q_3, \dots, q_N$. Take a point P in the field of these charges having distances $r_1, r_2, r_3, \dots, r_N$.

The potential V due to collection of N-point charges can be calculated by superposition principle which is stated as "the potential at a point due to one of the charges is not affected by the presence of the other charges"

The value of potential is given as

$$V = V_1 + V_2 + V_3 + \dots + V_N$$

$$V = \sum_{i=1}^N V_i$$

$$V = \sum_{i=1}^N \frac{kq_i}{r_i}$$

Q.5 Explain how electric potential can be calculated from electric field.

CALCULATION OF ELECTRIC POTENTIAL FROM ELECTRIC FIELD

Consider a $+q_0$ test charge is placed in

non uniform electric field $\vec{E}$ as shown in fig. Take a non linear path ab in this electric field.

The electric force $\vec{F}_e = q_0 \vec{E}$ acts on test charge $+q_0$ and moved it through displacement $d\vec{s}$. The amount of work done dW is given as

$$dW_{ab} = \vec{F}_e \cdot d\vec{s}$$

$$dW_{ab} = q_0 \vec{E} \cdot d\vec{s}$$

The net work done is

$$W_{ab} = \int_a^b q_0 \vec{E} \cdot d\vec{s} \quad \text{----- (1)}$$

The electric potential difference in terms of electric potential energy is defined as

$$\Delta V = \frac{\Delta U}{q_0} \quad \text{----- (2)}$$

The difference in potential energy ΔU , when q_0 moves between point "a" and "b" under the action of electric force $\vec{F}_e = q_0 \vec{E}$ is equal to negative work done and given as

$$\Delta U = -W_{ab}$$

Hence, we can write eq (2) as

$$\Delta V = -\frac{W_{ab}}{q_0} \quad \text{----- (3)}$$

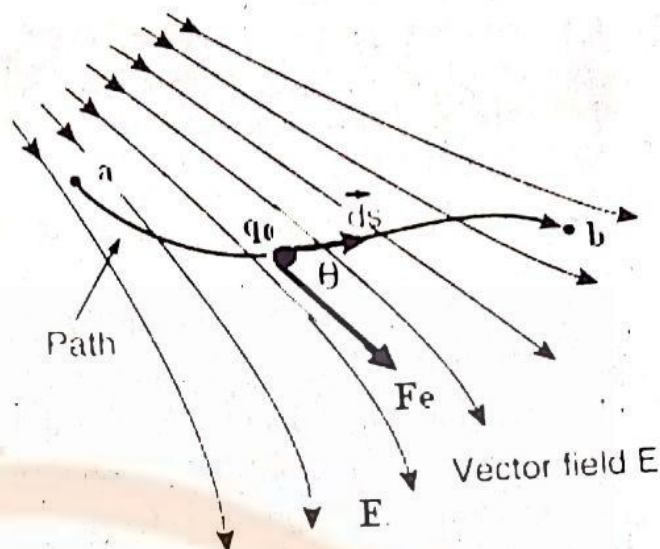
Put eq (1) in eq (3)

$$\Delta V = -\int_a^b \vec{E} \cdot d\vec{s}$$

$$V_b - V_a = -\int_a^b \vec{E} \cdot d\vec{s}$$

If we choose point "a" as reference point at infinity for which $V_a = 0$. The electric potential at point b is given as

$$V_b = -\int_a^b \vec{E} \cdot d\vec{s}$$



This formula is used to calculate electric potential difference between any two points in given electric field. It can also be used to calculate electric potential at any point in the given electric field.

Q.6 What is electric dipole? Calculate electric potential due to dipole at a point having distance r .

ELECTRIC DIPOLE

Two equal charges of opposite sign ($\pm q$) separated by distance d is called electric dipole.

ELECTRIC DIPOLE MOMENT

The product of magnitude of either charge and separation between charges is called dipole moment and denoted by $P = q d$

It is a vector quantity. Its direction is from negative charge towards positive charge. The unit of dipole moment is C-m.

ELECTRIC POTENTIAL DUE TO DIPOLE

Consider a $+q$ charge and $-q$ charge separated by distance d placed on z-axis as shown in fig. Take a point P having distance r_1 from $+q$ charge and r_2 from $-q$ charge.

The electric potential at P due to $+q$ charge is

$$V_1 = \frac{kq}{r_1}$$

The electric potential at P due to $-q$ charge is

$$V_2 = -\frac{kq}{r_2}$$

The net electric potential at P due to dipole is

$$V = V_1 + V_2$$

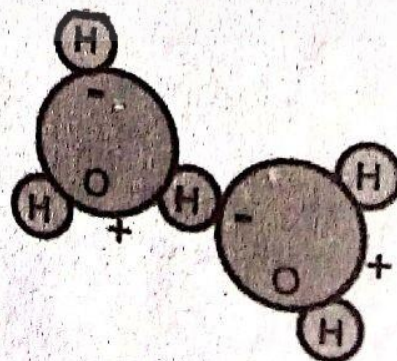
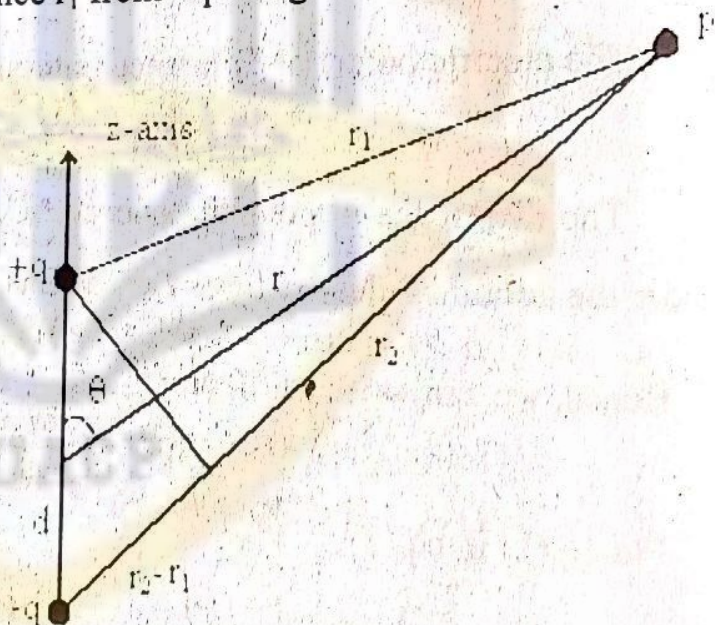
$$V = \frac{kq}{r_1} - \frac{kq}{r_2}$$

$$V = kq \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$V = kq \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

From fig. $r_2 - r_1 = d \cos \theta$

$$V = kq \left(\frac{d \cos \theta}{r_1 r_2} \right)$$



Consider naturally occurring dipoles such as water molecules as shown in diagram. Treating this system like a negative charge of 10 electrons and a positive charge of 10e, the effective separation of the negative and positive charge centers is 3.9×10^{-12} m. The point P is located very far from dipole such that $r \gg d$.

Under these conditions $r_1 r_2 = r^2$

$$V = k q \left(\frac{d \cos \theta}{r^2} \right)$$

$$V = k \left(\frac{P \cos \theta}{r^2} \right) \quad [P = q d]$$

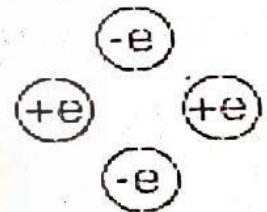
This is potential due to dipole.

Q.7 Calculate electric potential due to quadrupole?

QUADRUPOLE

The two electric dipoles arranged in such a way that they almost cancel electric effects of each other at distant points is called quadrupole. An elementary quadrupole can be represented as two dipoles oriented antiparallel.

A linear electric quadrupole can be created by superimposing two electric dipoles of opposite orientation so that their negative charges overlap. The most important uses of quadrupole is the characterization of nuclei.



ELECTRIC POTENTIAL DUE TO QUADRUPOLE

Consider two charges each of value $-q$ are placed at origin on x-axis. Now place $+q$ charge at point A having distance d from $-2q$. Similarly place an other $+q$ charge at point B having distance d from $-2q$ as shown in fig.

Now take a point P on z-axis having distance r from $-2q$ charge and distance $(r + d)$ from $+q$ charge at B. The distance of point P from $+q$ charge at A is $(r - d)$.

The electric potential at P due to charge $+q$ at A is

$$V_1 = \frac{k q}{r - d}$$

The electric potential at P due to $+q$ charge at B is

$$V_2 = \frac{k q}{r + d}$$

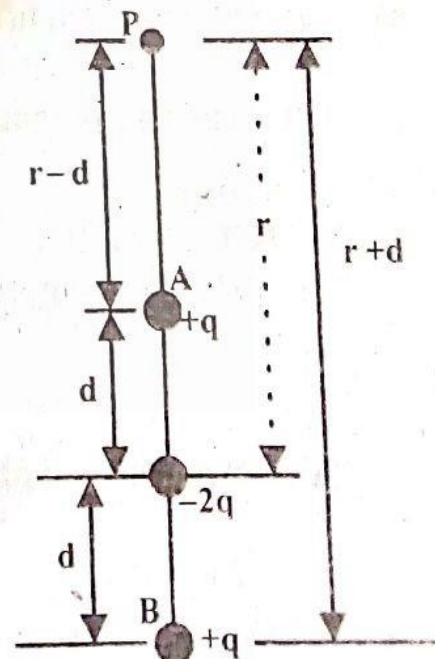
The electric potential at P due to $-2q$ charge

$$V_3 = \frac{k (-2q)}{r}$$

The net electric potential at P due to quadrupole is

$$V = V_1 + V_2 + V_3$$

$$V = \frac{k q}{r - d} + \frac{k q}{r + d} - \frac{k (2 q)}{r}$$



$$V = kq \frac{r(r+d) + r(r-d) - 2(r-d)(r+d)}{r(r-d)(r+d)}$$

$$V = kq \frac{r^2 + rd + r^2 - rd - 2(r^2 - d^2)}{r(r-d)(r+d)}$$

$$V = kq \frac{r^2 + rd + r^2 - rd - 2r^2 + 2d^2}{r(r-d)(r+d)}$$

$$V = \frac{k(2q d^2)}{r(r^2 - d^2)}$$

$$V = \frac{k(Q)}{r^3 \left(1 - \frac{d^2}{r^2}\right)}$$

Where $Q = 2q d^2$ is called electric quadrupole moment.

$$V = \frac{kQ}{r^3} \left(1 - \frac{d^2}{r^2}\right)^{-1/2}$$

Expand the term $\left(1 - \frac{d^2}{r^2}\right)^{-1/2}$ by binomial theorem

$$V = \frac{kQ}{r^3} \left(1 - \left(\frac{-1}{2}\right) \left(\frac{d^2}{r^2}\right) + \dots\right)$$

The term $\left(\frac{d^2}{r^2}\right) = 0$ because $r \gg d$. Therefore, neglect all other higher order terms.

The electric potential due to quadrupole is written as $V = \frac{kQ}{r^3}$

- ① The electric potential due to point charge varies as $\frac{1}{r}$ term called monopole term.
- ② The electric potential due to dipole varies as $\frac{1}{r^2}$ term called dipole term.
- ③ The electric potential due to quadrupole varies as $\frac{1}{r^3}$ term called quadrupole term and so on.

ELECTRIC FIELD

The electric potential at a distant point due to quadrupole is written as

$$V = \frac{kQ}{r^3} = kQ r^{-3}$$

The electric field and potential gradient are related as $E = -\frac{dV}{dr}$

$$E = -\frac{d}{dr} kQ r^{-3} = 3kQ r^{-4}$$

$$E = \frac{3kQ}{r^4}$$

Q.8 What is continuous charge distribution? Derive an expression to calculate electric potential at a point due to continuous charge distribution?

ELECTRIC POTENTIAL DUE TO CONTINUOUS CHARGE DISTRIBUTION

The electric charge is quantized. The collection of large number of charges is called continuous charge distribution. The continuous charge distribution has three types.

- ① Linear charge distribution
- ② Surface charge distribution
- ③ Volume charge distribution.

ELECTRIC POTENTIAL DUE TO LINEAR CHARGE DISTRIBUTION

Consider a linear charge distribution. Take a small length element dz of this distribution having charge dq .

dz length element has charge = dq

Unit length has charge = $\frac{dq}{dz}$

The charge per unit length is called linear charge density and denoted by λ

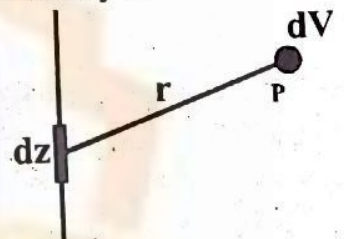
$$\lambda = \frac{dq}{dz}$$

The electric potential at P due to charge dq is

$$dV = \frac{k dq}{r} = \frac{k \lambda dz}{r}$$

The net electric potential at P due to whole linear charge distribution is

$$V = \int \frac{k \lambda}{r} dz$$



ELECTRIC POTENTIAL DUE TO SURFACE CHARGE DISTRIBUTION

Consider a surface charge distribution. Take a small area element da of this distribution having charge dq .

da area element has charge = dq

Unit area element has charge = $\frac{dq}{da}$

The charge per unit area is called surface charge density and denoted by σ

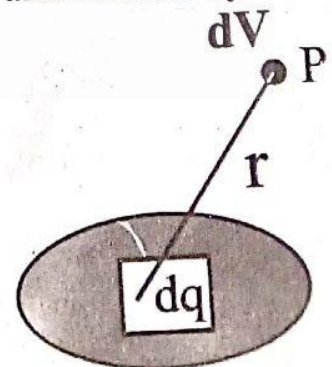
$$\sigma = \frac{dq}{da}$$

The electric potential at P due to charge dq is

$$dV = \frac{k dq}{r} = \frac{k \sigma da}{r}$$

The net electric potential at P due to whole

surface charge distribution is $V = \int \frac{k \sigma da}{r}$



ELECTRIC POTENTIAL DUE TO VOLUME CHARGE DISTRIBUTION

Consider a volume charge distribution. Take a small volume element dV_0 of this distribution having charge dq .

dV_0 volume element has charge $= dq$

Unit area element has charge $= \frac{dq}{dV_0}$

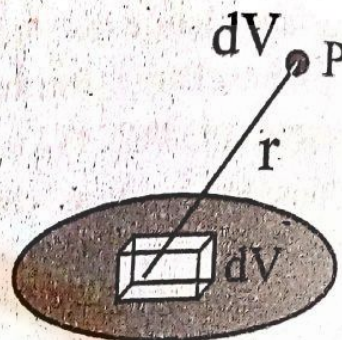
The charge per unit volume is called volume charge density and denoted by ρ

$$\rho = \frac{dq}{dV_0}$$

The electric potential at P due to charge dq is

$$dV = \frac{k dq}{r}$$

$$dV = \frac{k \rho dV_0}{r}$$



The net electric potential at P due to whole volume charge distribution is

$$V = \int \frac{k \rho}{r} dV_0$$

Q.9 Calculate electric potential at a point having distance z along the axis of ring of charge?

ELECTRIC POTENTIAL DUE TO RING OF CHARGE

Consider a uniformly charged ring having radius R . Take small length element ds of ring having charge dq . The linear charge density is defined as

$$\lambda = \frac{dq}{ds}$$

$$dq = \lambda ds$$

$$q = \lambda \int ds = \lambda(2\pi R)$$

Now take a point P on z -axis having distance z from the plane of ring. Electric potential dV due to charge dq at point P having distance r is

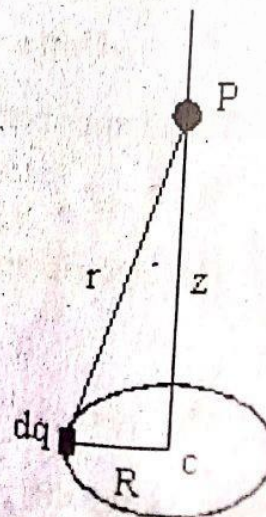
$$dV = \frac{k dq}{r}$$

$$dV = \frac{k \lambda ds}{r}$$

The net potential at P due to entire charge distribution of ring is

$$V = \int \frac{k \lambda ds}{r}$$

From fig. $r^2 = R^2 + z^2$



$$V = \int \frac{k \lambda ds}{[R^2 + z^2]^{1/2}} = \frac{k \lambda}{\sqrt{R^2 + z^2}} \int ds$$

$$V = \frac{k \lambda}{\sqrt{R^2 + z^2}} (2\pi R)$$

$$V = \frac{k q}{\sqrt{R^2 + z^2}}$$

This is potential at point P lying on axis of ring having distance z from the plane of the ring. The electric field at point P due to charge of ring is

$$E = -\frac{dV}{dz} = -\frac{d}{dz} \frac{k q}{\sqrt{R^2 + z^2}} = -\frac{d}{dz} k q (R^2 + z^2)^{-1/2}$$

$$E = -k q \left(-\frac{1}{2}\right) (R^2 + z^2)^{-3/2} (2z)$$

$$E = \frac{z k q}{(z^2 + R^2)^{3/2}}$$

Q.10 Calculate electric potential due to disk of charge?

ELECTRIC POTENTIAL DUE TO DISK OF CHARGE

Consider a plastic disk of radius R having uniform charge density σ on its top surface. Take a point P on the central axis of disk having distance z . The disk is combination of small flat rings. Now take such a ring having radius ω and width $d\omega$. The charge per unit area of ring is called surface charge density and written as

$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da$$

$$dq = \sigma(2\pi\omega)d\omega$$

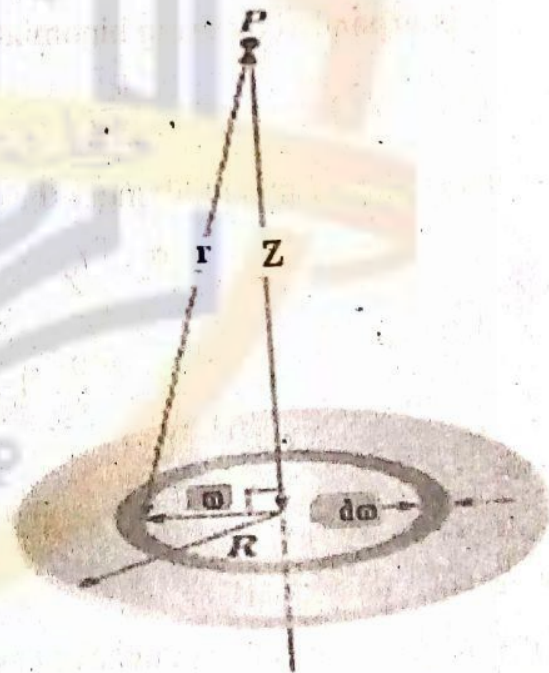
The electric potential at point P due to charge dq of ring having distance r is

$$dV = \frac{k dq}{r}$$

$$dV = \frac{k \sigma(2\pi\omega) d\omega}{\sqrt{\omega^2 + Z^2}} = \frac{1}{4\pi\epsilon_0} \frac{\sigma(2\pi\omega) d\omega}{\sqrt{\omega^2 + Z^2}}$$

The net potential at point P due to charged disk having radius R is

$$V = \int dV$$



$$V = \frac{1}{4\pi\epsilon_0} \int_0^R \frac{\sigma(2\pi\omega) d\omega}{\sqrt{\omega^2 + Z^2}}$$

$$V = \frac{\sigma}{2\epsilon_0} \int_0^R \frac{\omega d\omega}{\sqrt{\omega^2 + Z^2}}$$

$$V = \frac{\sigma}{2\epsilon_0} \int_0^R \frac{1}{2} (\omega^2 + Z^2)^{-1/2} (2\omega) d\omega$$

$$V = \frac{2\sigma}{4\epsilon_0} \left[(\omega^2 + Z^2)^{1/2} \right]_0^R$$

$$V = \frac{\sigma}{2\epsilon_0} [(R^2 + Z^2)^{1/2} - Z]$$

In special case $Z \gg R$

$$(R^2 + Z^2)^{1/2} = Z \left(1 + \frac{R^2}{Z^2} \right)^{1/2} = Z \left(1 + \frac{1}{2} \frac{R^2}{Z^2} + \dots \right)$$

It is expanded by using binomial theorem. Neglect the higher order terms we get

$$(R^2 + Z^2)^{1/2} = Z + \frac{R^2}{2Z}$$

The value of potential under these conditions is

$$V = \frac{\sigma}{2\epsilon_0} \left[Z + \frac{R^2}{2Z} - Z \right]$$

$$V = \frac{\sigma}{2\epsilon_0} \left(\frac{R^2}{2Z} \right) = \frac{\sigma(\pi R^2)}{4\pi\epsilon_0 Z}$$

$$V = \frac{kq}{Z}$$

$(\pi R^2) R = \text{surface area of disc}$

Where $q = \int \sigma da = \sigma(\pi R^2)$

This shows that charged disk behaves as point charge in limiting case when $Z \gg R$

Q.11 What are equipotential surfaces? Draw the lines of force and cross-sections of equipotential surfaces for (i) a uniform field (ii) a positive point charge.

EQUIPOTENTIAL SURFACES

The family of surfaces that connect points having the same value of the electric potential are called equipotential surfaces.

Consider uniform electric field E indicated by horizontal electric lines of force. The perpendicular dashed lines are cross-sections of equipotential surfaces. Take two points A and B having distance L

The potential difference between these two points is given as

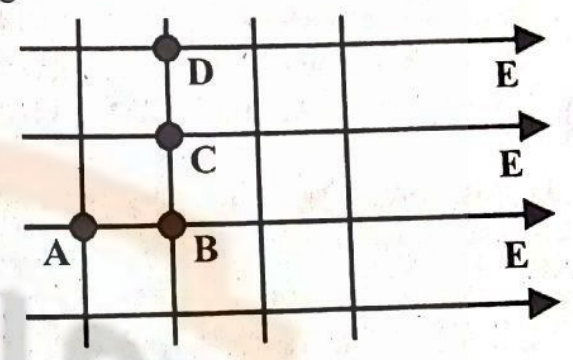
$$\Delta V = - \int_A^B \vec{E} \cdot d\vec{S} = -EL$$

The magnitude of work done by electric field on unit positive test charge q_0 when moved from A to B is $q_0 EL$

The potential difference between B and C is given as

$$\Delta V = - \int_B^C \vec{E} \cdot d\vec{S}$$

$$\Delta V = - \int_B^C E dS \cos 90^\circ$$



This relation can be written as

$$V_C - V_B = 0$$

$$V_C = V_B$$

Since $\Delta V = -\frac{W_{BC}}{q_0} = 0$ so $W_{BC} = 0$. Hence no work is done by electric field when q_0 is moved from B to C.

In this way all the points on the line containing points B, C, D etc have the same potential. The surfaces passing through these points are called equipotential surfaces.

Consider a $+q$ charge is placed at the center of a sphere. The electric field lines are along radius direction. The potential difference between two points is given as

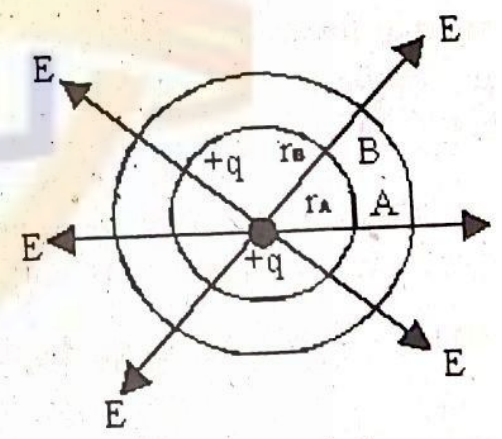
$$V_A - V_B = kq \left(\frac{1}{r_A} - \frac{1}{r_B} \right)$$

When points A and B lies on the surface of same sphere $r_A = r_B$. In this case

$$V_A - V_B = kq \left(\frac{1}{r_A} - \frac{1}{r_A} \right)$$

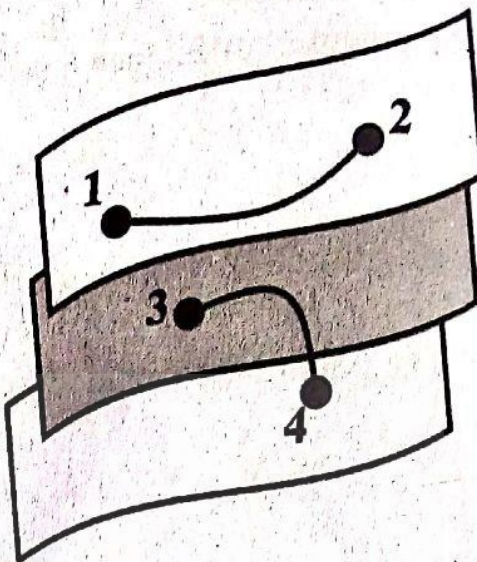
$$V_A - V_B = 0$$

$$V_A = V_B$$



It means all points at a given radius have the same potential. Therefore, the equipotential surfaces of a point charge form a family of concentric spheres. For a dipole the equipotential surfaces are more complicated.

The fig shows the family of equipotential surfaces. It has following features.



- ① The work done by the electric field when a charge moves along point 1 to point 2 is zero because both paths end and start at same surface.
- ② Along point 3 to 4, work done is non zero because path starts from one surface and ends at other surface. The equipotential surfaces are always right angles to the electric lines of force.

Q.12 Explain how electric field is calculated from given electric potential or prove that negative of potential gradient is equal to electric field ($E = -\text{grad}V$)

ELECTRIC FIELD AND POTENTIAL GRADIENT ($E = -\text{grad} V$)

The maximum value of rate of increase in electric potential in a particular direction is called potential gradient.

Consider a cross section of a family of equipotential surfaces which differ in potential by amount dV as shown in fig. The electric field E at a particular point P is at right angle to equipotential surface passing through P . Now place a test charge $+q_0$ at point P having potential V .

The amount of work done dW by electric field $\vec{E}$ on test charge $+q_0$ when it is moved from equipotential surface having potential V to equipotential surface having potential $(V + dV)$ through displacement $\vec{ds}$ is

$$dW = -q_0 dV$$

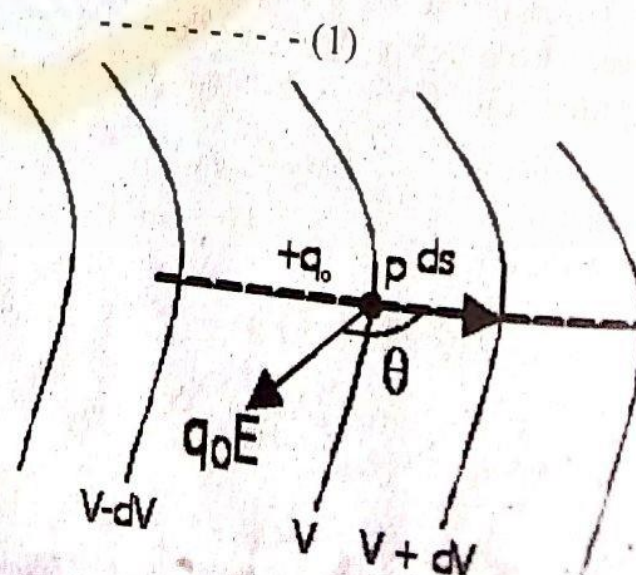
The amount of work done by electric field $\vec{E}$ on charge $+q_0$ in terms of displacement $\vec{ds}$ is

$$dW = \vec{F} \cdot \vec{ds}$$

$$dW = q_0 \vec{E} \cdot \vec{ds}$$

$$dW = q_0 E ds \cos\theta \quad \text{----- (2)}$$

Comparing eq (1) and eq (2)



$$q_0 E \cos\theta = -q_0 dV$$

$$E \cos\theta = -\frac{dV}{ds}$$

This equation states that negative rate of change of potential with position in any direction is equal to component of E in that direction. The negative sign means E points in the direction of decreasing V .

Take maximum value on both sides

$$(E \cos\theta)_{\max} = -\left(\frac{dV}{ds}\right)_{\max}$$

The maximum value of $\cos\theta$ is one

$$E = -\left(\frac{dV}{ds}\right)_{\max} \quad \text{-----(3)}$$

The maximum value of $\frac{dV}{ds}$ at a point is called potential gradient at that point.

$$\text{grad } V = \left(\frac{dV}{ds}\right)_{\max} \quad \text{-----(4)}$$

Comparing eq(3) and eq(4)

$$E = -\text{grad } V$$

In vector form this relation can be written as

$$\vec{E} = -\vec{\nabla} V$$

In component form we can write as

$$E_x \hat{i} + E_y \hat{j} + E_z \hat{k} = -\left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}\right) V$$

On comparing coefficients of unit vectors

$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}$$

SHORT QUESTIONS

Q:1 What would happen to you if you were on an insulated stand and your potential was increased by 10 kV with respect to the Earth?

Ans: The current will not flow through you when you were on an insulated stand and your potential was increased by 10 kV with respect to the Earth because circuit is not completed.

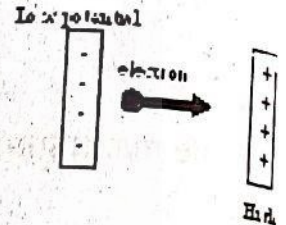
Q:2 Would a proton volt compare with an electron volt. The mass of a proton is 1840 times that of an electron?

Ans: It makes no difference either we use proton volt or electron volt because electron volt = $e V = \text{charge of electron} \times 1 V$. The charge of electron and charge of proton is same in magnitude.

Q:3 Do electrons tend to go to region of high potential or of low potential?

Ans: The electron tends to go from low potential region to high potential region.

The low potential (-) side pushes negatively charged electron and high potential (+) side attracts negatively charged electron.



Q:4 Does the amount of work per unit charge required to transfer electric charge from one point to another point in electric field depends upon the amount of charge transferred?

Ans: We know that $\Delta V = W/q_0$. Therefore, the amount of work per unit charge required to transfer electric charge from one point to another point in electric field depends upon the amount of charge transferred?

Q:5 Distinguish between potential difference and potential energy?

Ans: The amount of work per unit charge required to transfer electric charge from one point to another point against electric field keeping electrostatic equilibrium is called potential difference. It is written as $\Delta V = W/q_0$. Its SI unit is called volt. When a charge is transferred from one point to another point against electric field keeping electrostatic equilibrium, work is done. This work is stored in charge as potential energy. It is written as $\Delta U = -W$. Its SI unit is called Joule.

Q:6 Can there be a potential difference between two conductors that carry like charges of the same magnitude.

Ans: There will be no potential difference between two conductors that carry like charges of the same magnitude.

Q:7 Give examples of situations in which the potential of a charged body has a sign opposite to that of its charge.

Ans: Consider a point P having distance r from a $+q$ charge. The potential at P due to $+q$ charge is

$$V_p = \frac{kq}{r}$$

The potential at infinity due to $+q$ charge is

$$V_\infty = \frac{kq}{r_\infty} = 0$$

The change in potential

$$V_\infty - V_p = -\frac{kq}{r}$$

Q:8 Can two different equipotential surfaces intersect

Ans: The surface whose every point has same potential is called equipotential surface. When two equipotential surfaces intersect, the point of intersection will have different potentials. It means one surface has different potentials. This is not possible. Hence two different equipotential surfaces never intersect.

Q:9 If E equal to zero at a given point, must V also equal to zero for that point.

Ans: It is given that, $E = 0$. The electric field and potential are related as

$$E = -\frac{dV}{dr}$$

$$0 = -\frac{dV}{dr}$$

$$dV = 0$$

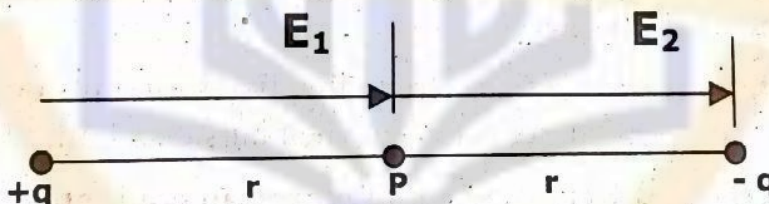
$$V = \text{constant}$$

It means if E is equal to zero at a given point, it is not necessary that V also equal to zero for that point.

Q:10 If V equal to zero at a given point, must E also equal to zero for that point. Give an example to explain your answer.

Ans: If V equal to zero at a given point, It is not necessary that E must also equal to zero for that point.

Consider a $+q$ charge and $-q$ charge are separated by distance $2r$ as shown in fig.



The electric field at mid point P due to $+q$ charge is $E_1 = \frac{kq}{r^2}$. The electric field at

mid point P due to $-q$ charge is $E_2 = \frac{kq}{r^2}$.

The net electric field at P is

$$E = E_1 + E_2 = \frac{2kq}{r^2}$$

The electric potential at mid point P due to $+q$ charge is $V_1 = \frac{kq}{r}$. The electric

potential at mid point P due to $-q$ charge is $V_2 = -\frac{kq}{r}$. The net electric potential

at P is

$$V = V_1 + V_2 = 0$$

Q:11 If you know E only at a given point, can you calculate V at that point.
Ans: If you know E only at a given point, then you can calculate V at that point by using following relation.

$$E = -\frac{dV}{dr}$$

For example if $V = \frac{kq}{r}$, then by above relation electric field $E = \frac{kq}{r^2}$

Q:12 Is the uniformly charged non-conducting disk is a surface of constant potential.

Ans: The electric potential at the center of disk having radius R is given as

$$V = \frac{\sigma R}{2\epsilon_0} = \text{Constant}$$

Therefore, uniformly charged non-conducting disk is a surface of constant potential.

Q:13 We have seen that inside a hollow conductor, you are shielded from the fields of outside charges. If you are outside a hollow conductor that contains charges, are you shielded from the fields of these charges.

Ans: The shell theorem states that all the charge of a charged shell is concentrated at center of shell for points outside the shell and electric field is zero for points inside the charged shell.

Therefore, we are shielded from the fields of outside charges when inside a hollow conductor but not shielded when outside a hollow charged conductor.

Q:14 If the surface of a charged conductor is an equipotential, does that mean that charge is distributed uniformly over that surface. If the electric field is constant in magnitude over the surface of a charged conductor, does that mean that charge is distributed uniformly?

Ans: Yes, if the surface of a charged conductor is an equipotential, it means that charge is distributed uniformly over that surface.

Yes, if the electric field is constant in magnitude over the surface of a charged conductor, it means that charge is distributed uniformly.

Q:15 Why can an isolated atom not have a permanent electric dipole moment.

Ans: The pair of $+q$ charge and $-q$ charge having constant distance is called dipole. The dipole moment is defined as $P = qd$. The electrons are in continuous motion in different orbits. Therefore $-q$ charge (electrons) does not have constant distance from $+q$ charge (nucleus). That is why an isolated atom can not have a permanent electric dipole moment.

Q:16 If V is a constant through out a given region of space, what can you say about E in that region.

Ans: It is given that $V = \text{constant}$

The electric field and potential are related as

$$E = -\frac{dV}{dr}$$

$$E = -\frac{d(\text{constant})}{dr}$$

$$E = 0$$

If V is a constant through out a given region of space, then you can say $E = 0$ in that region.

Q:17 How can you ensure that the electric potential in a given region of space will have a constant value.

Ans: The electric field and potential are related as

$$E = -\frac{dV}{dr}$$

If $E = 0$,

$$0 = -\frac{dV}{dr}$$

$$dV = 0$$

$$V = \text{constant}$$

If $E = 0$ in some region, then V will be constant through out that given region of space.

Q:18 Device an arrangement of three point charges separated by finite distances that has zero electric potential energy.

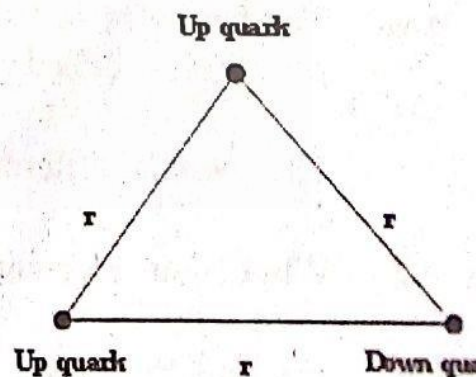
Ans: Consider two up quarks having charge $\frac{+2e}{3}$ and

down quark having charge $\frac{-e}{3}$ are placed at corners of equilateral triangle.

$$U = U_1 + U_2 + U_3$$

$$U = \frac{k\left(\frac{2e}{3}\right)\left(\frac{2e}{3}\right)}{r} + \frac{k\left(\frac{2e}{3}\right)\left(\frac{-e}{3}\right)}{r} + \frac{k\left(\frac{2e}{3}\right)\left(\frac{-e}{3}\right)}{r}$$

$$U = 0$$



Q:19 An electrical worker was accidentally electrocuted and a newspaper account reported. He accidentally touches a high voltage cable and 20000 V of electricity surged through his body. Criticize this statement.

Ans: The statement of newspaper must be as "He accidentally touches a high voltage cable and 20000 V of voltage surged through his body".

Q:20 What is the effect on measured values of potential difference, if we assume that potential of earth is +100 V or -100 V instead of zero.

Ans: The earth is taken as a reference of potential and assigned the value of zero potential. This is taken just for convenience. The other choice would change the potential everywhere by the same amount but would not change the results for the potential difference.

Q:21 Why the electron volt is more convenient unit of energy than the joules.

Ans: The unit eV for energy is a great convenience when dealing with atoms or nuclei in which the charge is easily expressed in terms of e.

Q:22 The electric potential is a scalar quantity. What can you say about its gradient?

Ans: The electric potential is a scalar quantity but its gradient is a vector quantity because gradient of a scalar function is a vector quantity.

Q:23 Define equipotential surfaces. What is potential difference between two points on a equipotential surfaces?

Ans. The family of surfaces that connect points having the same value of the electric potential are called equipotential surfaces.

The potential difference between two points on a equipotential surfaces is zero.

Q:24 What is one electron volt. Find its value in joules.

Ans. The unit of energy is called electron volt. The energy gained by or lost by an electron when it moves in a potential difference of one volt is called electron volt.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

SAMPLE PROBLEMS

- P.1 Two protons in nucleus of U^{238} atom are 6.0 fm apart. What is potential energy associated with the electric force that acts between these two particles?

Sol.

DATA

$$q_1 = 1.6 \times 10^{-19} \text{ C}$$

$$q_2 = 1.6 \times 10^{-19} \text{ C}$$

$$r = 6 \times 10^{-15} \text{ m}$$

$$U = ?$$

CALCULATIONS

$$U = \frac{kq_1q_2}{r}$$

$$U = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{6 \times 10^{-15}}$$

$$U = 3.8 \times 10^{-14} \text{ J} = 2.4 \times 10^5 \text{ eV}$$

- P.2 Calculate the electric potential energy of a system of three charges $q_1 = q$, $q_2 = -4q$ and $q_3 = +2q$. Where $q = 150 \text{ nC}$ placed at the vertices of an equilateral triangle having each side 12 cm.

Sol.

DATA

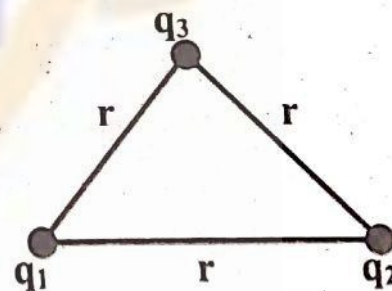
$$q_1 = q$$

$$q_2 = -4q$$

$$q_3 = +2q$$

$$q = 150 \text{ nC} = 150 \times 10^{-9} \text{ C}$$

$$r = 12 \text{ cm} = 12 \times 10^{-2} \text{ m}$$



CALCULATIONS

$$U = \frac{kq_1q_2}{r} + \frac{kq_2q_3}{r} + \frac{kq_3q_1}{r}$$

$$U = -\frac{4kq^2}{r} - \frac{8kq^2}{r} + \frac{2kq^2}{r}$$

$$U = -\frac{10kq^2}{r} = -\frac{10 \times 9 \times 10^9 (150 \times 10^{-9})^2}{12 \times 10^{-2}}$$

$$U = 1.68 \times 10^{-2} \text{ J}$$

P.3 An alpha particle in a nuclear accelerator moves from one terminal of a potential of $V_a = 6.5 \times 10^6 \text{ V}$ to another at a potential of $V_b = 0$. (a) What is the corresponding change in potential energy of the system. (b) What is change in KE of the particle

Sol.

DATA

$$q = 2e = 2 \times 1.6 \times 10^{-19} \text{ C}$$

$$V_a = 6.5 \times 10^6 \text{ V}$$

$$V_b = 0$$

CALCULATIONS

$$\begin{aligned} \text{(a)} \quad \Delta U &= q(V_b - V_a) \\ \Delta U &= 2 \times 1.6 \times 10^{-19} (0 - 6.5 \times 10^6) \\ \Delta U &= -2.1 \times 10^{-12} \text{ J} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \Delta \text{KE} &= -\Delta U \\ \Delta \text{KE} &= +2.1 \times 10^{-12} \text{ J} \end{aligned}$$

P.4 What must be the magnitude of an isolated +ve point charge from the electric potential at 15 cm from the charge to be 120 V?

Sol.

DATA

$$V = 120 \text{ V}$$

$$r = 15 \times 10^{-2} \text{ m}$$

$$q = ?$$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{Vr}{k} = \frac{120 \times 15 \times 10^{-2}}{9 \times 10^9}$$

$$q = 2 \times 10^{-9} \text{ C}$$

P.5 What is the electric potential at the surface of a gold nucleus? The radius is $7 \times 10^{-15} \text{ m}$ and atomic no. = 79?

Sol.

DATA

$$q = 79e = 79 \times 1.6 \times 10^{-19} \text{ C}$$

$$r = 7 \times 10^{-15} \text{ m}$$

$$V = ?$$

$$V = \frac{kq}{r}$$

$$V = \frac{9 \times 10^9 \times 79 \times 1.6 \times 10^{-19}}{7 \times 10^{-15}}$$

$$V = 1.6 \times 10^7 \text{ Volt}$$

P.6 Four charges $q_1 = 12 \text{ nC}$, $q_2 = -24 \text{ nC}$, $q_3 = 31 \text{ nC}$ and $q_4 = 17 \text{ nC}$ are placed at four corners of a square having each side $d = \sqrt{2} \text{ m}$. Calculate the resultant potential at the center of square.

Sol.

DATA

$$q_1 = 12 \text{ nC} = 12 \times 10^{-9} \text{ C}$$

$$q_2 = -24 \text{ nC} = -24 \times 10^{-9} \text{ C}$$

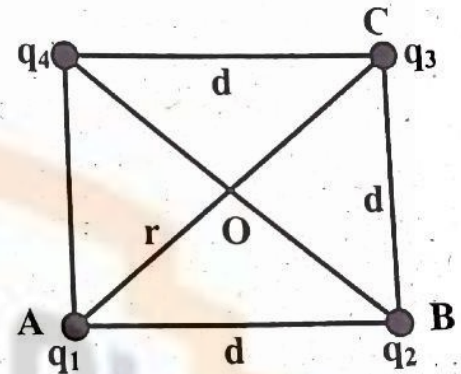
$$q_3 = 31 \text{ nC} = 31 \times 10^{-9} \text{ C}$$

$$q_4 = 17 \text{ nC} = 17 \times 10^{-9} \text{ C}$$

$$d = \sqrt{2} \text{ m} = 1.41 \text{ m}$$

$$d^2 = 2 \text{ m}$$

$$V = ?$$



CALCULATIONS

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = d^2 + d^2 = 4$$

$$AC = 2$$

$$r = \frac{AC}{2} = \frac{2}{2} = 1 \text{ m}$$

The resultant potential at the center of square

$$V = \frac{kq_1}{r} + \frac{kq_2}{r} + \frac{kq_3}{r} + \frac{kq_4}{r}$$

$$V = \frac{k}{r} (q_1 + q_2 + q_3 + q_4)$$

$$V = \frac{9 \times 10^9}{1} (12 - 24 + 31 + 17) \times 10^{-9}$$

$$V = 324 \text{ Volt}$$

P.7 The potential at P on the axis of a circular plastic disk of radius R, one surface of which carries uniform charge density σ is: $V = \frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + Z^2} - Z)$
Derive an expression for electric field at axial points?

Sol.

DATA

$$V = \frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + Z^2} - Z)$$

$$E = ?$$

CALCULATIONS

$$V = \frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + Z^2} - Z)$$

Differentiate with respect to z.

$$\frac{\partial V}{\partial Z} = \frac{\sigma}{2\epsilon_0} \left(\frac{1}{2} (R^2 + Z^2)^{-1/2} (2Z) - 1 \right)$$

$$\frac{\partial V}{\partial Z} = \frac{-\sigma}{2\epsilon_0} \left(1 - \frac{Z}{(R^2 + Z^2)^{1/2}} \right)$$

The electric field at axial point is

$$E = -\frac{\partial V}{\partial Z}$$

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{Z}{(R^2 + Z^2)^{1/2}} \right)$$

EXERCISE PROBLEMS

P.1 Consider a point charge $q = +15 \text{ nC}$ and take $V = 0$ at infinity. What is the radius of an equipotential surface having a potential of 30 V due to charge q alone.

Sol.

DATA

$$q = +15 \text{ nC} = 15 \times 10^{-9} \text{ C}$$

$$V = 30 \text{ V}$$

$$r = ?$$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$r = \frac{kq}{V} = \frac{9 \times 10^9 \times 15 \times 10^{-9}}{30} = 4.5 \text{ m}$$

- P.2 A point charge $q = +1.16 \mu\text{C}$ is located at origin. Consider point A which is 206 cm distant towards east from origin and point B which is 117 cm distant towards west from the charge. Find the potential difference $V_A - V_B$.
Sol.

DATA

$$q = +1.16 \mu\text{C} = 1.16 \times 10^{-6} \text{ C}$$

$$r_A = 206 \text{ cm} = 206 \times 10^{-2} \text{ m}$$

$$r_B = 117 \text{ cm} = 117 \times 10^{-2} \text{ m}$$

CALCULATIONS

$$V_A = \frac{kq}{r_A}$$

$$V_A = \frac{9 \times 10^9 \times 1.16 \times 10^{-6}}{206 \times 10^{-2}} = 5068 \text{ V}$$

$$V_B = \frac{kq}{r_B}$$

$$V_B = \frac{9 \times 10^9 \times 1.16 \times 10^{-6}}{117 \times 10^{-2}} = 8923 \text{ V}$$

The potential difference is

$$V_A - V_B = 5068 - 8923 = -3855 \text{ V}$$

- P.3 The electric Pd between discharge points during a particular thunderstorms is $1.23 \times 10^9 \text{ V}$. What is the magnitude of the change in electrical potential energy of an electron that moves between these points give your answer in (a) joule (b) electron volts.
Sol.

DATA

$$\Delta V = 1.23 \times 10^9 \text{ V}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$\Delta U = ?$$

CALCULATIONS

$$(a) \quad \Delta U = q \Delta V = 1.6 \times 10^{-19} \times 1.23 \times 10^9$$

$$\Delta U = 1.97 \times 10^{-10} \text{ J}$$

$$(b) \quad \Delta U = \frac{1.97 \times 10^{-10}}{1.6 \times 10^{-19}} = 1.23 \times 10^9 \text{ eV}$$

- ✓P.4 An infinite sheet of charge has density $\sigma = 0.12 \mu\text{C}/\text{m}^2$. How far apart are the equipotential surfaces whose potentials differ by 48 V.

Sol.

DATA

$$\sigma = 0.12 \times 10^{-6} \text{ C}/\text{m}^2$$

$$\Delta V = 48 \text{ V}$$

$$\Delta r = ?$$

CALCULATIONS

$$E = \frac{\sigma}{2\epsilon_0}$$

$$E = \frac{0.12 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} = 6779.6 \text{ N/C}$$

The apart between the equipotential surfaces

$$\Delta V = E\Delta r$$

$$\Delta r = \frac{\Delta V}{E} = \frac{48}{6779.6}$$

$$\Delta r = 7.08 \times 10^{-3} \text{ m}$$

- P.5** In the quarks model of fundamental particles, a proton is composed of three quarks two up quarks each having charge $+\frac{2e}{3}$ and one down quark having charge $-\frac{e}{3}$. Suppose that the three quarks are equidistant from each other. Take the distance to be $1.32 \times 10^{-15} \text{ m}$ and (a) calculate the potential energy of the interaction between two up quarks and (b) total electrical potential energy of the system?

Sol.

DATA

$$\text{Charge of down quark} = -\frac{e}{3}$$

$$\text{Charge of first up quark} = q_1 = +\frac{2e}{3}$$

$$\text{Charge of second up quark} = q_2 = +\frac{2e}{3}$$

$$\text{Distance between quarks} = r = 1.32 \times 10^{-15} \text{ m}$$

$$\text{Potential Energy} = U = ?$$

CALCULATIONS

(a) For up quarks

$$U = \frac{kq_1q_2}{r} = \frac{k(2e)(2e)}{3(3)r}$$

$$U = \frac{4 \times 9 \times 10^9 (1.6 \times 10^{-19})^2}{9 \times 1.32 \times 10^{-15}}$$

$$U = 7.75 \times 10^{-14} \text{ J}$$

The potential energy between 1st up quark and down quark is

$$U_1 = \frac{kq_1q_2}{r} = \frac{-2ke^2}{9r}$$

The potential energy between 2nd up quark and down quark is

$$U_2 = \frac{kq_1q_2}{r} = \frac{-2ke^2}{9r}$$

The potential energy between 1st up quark and 2nd up quark is

$$U_3 = \frac{kq_1q_2}{r} = \frac{4ke^2}{9r}$$

(b) Total energy = $U_1 + U_2 + U_3 = 0$

- P.6 Two parallel plat conducting surfaces of spacing $d = 1.0$ cm have a potential difference ΔV of 10.3 kV. An electron is projected from one plate directly towards second. What is the initial velocity of electron if it comes to rest just at the surface second plate? Ignore the relativistic effect.

Sol.

DATA

$$d = 1 \times 10^{-2} \text{ m}$$

$$\Delta V = 10.3 \times 10^3 \text{ V}$$

$$q = -1.6 \times 10^{-19} \text{ C}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$V_f = 0$$

$$V_i = ?$$

CALCULATIONS:

From work energy relation

$$F d = \frac{mV_f^2}{2} - \frac{mV_i^2}{2}$$

$$q E d = \frac{mV_f^2}{2} - \frac{mV_i^2}{2} \quad \left[E = -\frac{\Delta V}{d} \right]$$

$$q E d = -\frac{mV_i^2}{2}$$

$$V_i^2 = \frac{2 q \Delta V}{m} = \frac{2 \times 1.6 \times 10^{-19} \times 10.3 \times 10^3}{9.1 \times 10^{-31}}$$

$$V_i = 6.0 \times 10^7 \text{ m/s}$$

- P.7. Two large parallel conducting plates are 12 cm apart and carry equal but opposite charge on their facing surfaces. An electron placed midway between the two plates experiences a force of $3.9 \times 10^{-15} \text{ N}$ (a) Find the electric field at the position of electron (b) What is the Pd between the plates?

Sol.

DATA

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$\Delta r = 12 \times 10^{-2} \text{ m}$$

$$F = 3.9 \times 10^{-15} \text{ N}$$

CALCULATIONS

$$(a) \quad E = \frac{F}{q} = \frac{3.9 \times 10^{-15}}{1.6 \times 10^{-19}} = 24375 \text{ N/C}$$

$$(b) \quad \Delta V = E \Delta r$$

$$\Delta V = 24375 \times 12 \times 10^{-2} = 2925 \text{ V}$$

P.8

In the Milliken oil drop experiment an electric field of $1.92 \times 10^5 \text{ N/C}$ is maintained at balance across two plates separated by 1.5 cm. Find ΔV between plates?

Sol.

DATA

$$E = 1.92 \times 10^5 \text{ N/C}$$

$$\Delta r = 1.5 \times 10^{-2} \text{ m}$$

$$\Delta V = ?$$

CALCULATIONS

$$\Delta V = E \Delta r$$

$$\Delta V = 1.92 \times 10^5 \times 1.5 \times 10^{-2}$$

$$\Delta V = 2880 \text{ Volts}$$

P.9

A gold nucleus contains a positive charge equal to that of 79 protons and has a radius of 7 fm. An alpha particle has a KE at point far from the nucleus and is traveling directly towards it. The alpha particle just touches the surface of the nucleus where its velocity is reversed in direction. Calculate KE.

Sol.

DATA

$$q_1 = 79 e$$

$$q_2 = 2 e$$

$$r = 7 \times 10^{-15} \text{ m}$$

$$\text{KE} = ?$$

CALCULATIONS

$$\text{K.E.} = \text{P.E.}$$

$$\text{K.E.} = \frac{k q_1 q_2}{r}$$

$$\text{KE} = \frac{9 \times 10^9 \times 79 \times 2 (1.6 \times 10^{-19})^2}{7 \times 10^{-15}}$$

$$\text{KE} = 5.2 \times 10^{-12} \text{ J}$$

- P.10** Compute the escape speed for an electron from the surface of a uniformly charged sphere of radius 1.22 cm and total charge 1.76×10^{-15} C. Neglect gravitational forces.

Sol.

DATA

$$\begin{aligned} e &= 1.6 \times 10^{-19} \text{ C} \\ q &= 1.76 \times 10^{-15} \text{ C} \\ r &= 1.22 \times 10^{-2} \text{ m} \\ V &= ? \end{aligned}$$

CALCULATIONS

$$\text{K.E.} = \frac{1}{2} mV^2$$

$$\text{K.E.} = eV_0$$

Comparing

$$eV_0 = \frac{1}{2} mV^2$$

$$e \frac{kq}{r} = \frac{1}{2} mV^2$$

$$V = \sqrt{\frac{2keq}{rm}}$$

$$V = \left(\frac{2 \times 9 \times 10^9 \times 1.76 \times 10^{-15} \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31} \times 1.22 \times 10^{-2}} \right)^{1/2}$$

$$V = 21367 \text{ m/s}$$

- P.11** A space shuttle moves through the dilute ionized gas of the earth ionosphere its potential is typically changes by -1.0 V before it complete one revolution. By assuming that the shuttle is a sphere of radius 10 cm, estimate the amount of charge it collects.

Sol.

DATA

$$\begin{aligned} r &= 10 \times 10^{-2} \text{ m} \\ V &= -1.0 \text{ Volt} \\ q &= ? \end{aligned}$$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{Vr}{k}$$

$$q = \frac{-1 \times 10 \times 10^{-2}}{9 \times 10^9} = -1.11 \times 10^{-11} \text{ C}$$

- P.12** An electric field of 100 V/m is often observed near the surface of the earth. If this field were the same over the entire surface, what would be the electric potential of a point on the surface? The radius of earth is 6400 km.

Sol.

DATA

$$E = 100 \text{ V/m}$$

$$\Delta r = 6.4 \times 10^6 \text{ m}$$

$$\Delta V = ?$$

CALCULATIONS

$$\Delta V = E \Delta r$$

$$\Delta V = 100 \times 6.4 \times 10^6 = 6.4 \times 10^8 \text{ Volts}$$

- P.13** The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47D where D is debye unit = $3.34 \times 10^{-30} \text{ C-m}$. Calculate the electric potential due to an ammonia molecule at point 52 nm away along the axis of dipole.

Sol.

DATA

$$P = 1.47\text{D} = 1.47 \times 3.34 \times 10^{-30} \text{ C-m}$$

$$r = 52 \times 10^{-9} \text{ m}$$

$$V = ?$$

CALCULATIONS

$$V = \frac{kP}{r^2} = \frac{9 \times 10^9 \times 1.47 \times 3.34 \times 10^{-30}}{(52 \times 10^{-9})^2}$$

$$V = 1.6 \times 10^{-5} \text{ V}$$

- P.14** Two large parallel metal plates are 1.48 cm apart and carry equal but opposite charges on their facing surface. The negative plate is grounded and its potential is taken to be zero. If the potential halfway between the plates is 5.12 V, what is the electric field in this region?

Sol.

DATA

$$S = 1.48 \times 10^{-2} \text{ m}$$

$$dV = 5.12 \text{ Volt}$$

$$E = ?$$

CALCULATIONS

$$dr = \frac{S}{2}$$

$$dr = \frac{1.48}{2} \times 10^{-2} = 7.4 \times 10^{-3} \text{ m}$$

$$E = \frac{dV}{dr} = \frac{5.12}{7.4 \times 10^{-3}} = 692 \text{ V/m}$$

P.15 From eq. $\frac{Kq}{(R^2 + Z^2)^{1/2}}$. Derive an expression for E at axial points uniformly charged ring.

Sol.

DATA

$$V = \frac{kq}{(R^2 + Z^2)^{1/2}} = kq(R^2 + Z^2)^{-1/2}$$

CALCULATIONS

$$V = kq(R^2 + Z^2)^{-1/2}$$

$$\frac{dV}{dr} = \frac{-kqZ}{(R^2 + Z^2)^{3/2}}$$

The electric field is

$$E = \frac{-dV}{dr}$$

$$E = \frac{+kqZ}{(R^2 + Z^2)^{3/2}}$$

P.16 Calculate the radial potential gradient in V/m at the surface of the gold nucleus of radius 7.0×10^{-15} m and 79 protons.

Sol.

DATA

$$r = 7.0 \times 10^{-15} \text{ m}$$

$$q = 79e = 79 \times 1.6 \times 10^{-19} \text{ C}$$

$$\frac{dV}{dr} = ?$$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$\frac{dV}{dr} = \frac{-kq}{r^2} = \frac{-9 \times 10^9 \times 79 \times 1.6 \times 10^{-19}}{(7.0 \times 10^{-15})^2}$$

$$\frac{dV}{dr} = -2.3 \times 10^{21} \text{ V/m}$$

P.17 A charge of 15 nC can be produced by simple rubbing. To what potential would such a charge raise an isolated conducting sphere of 16 cm radius.

Sol.

DATA

$$q = 15 \times 10^{-9} \text{ C}$$

$$r = 16 \times 10^{-2} \text{ m}$$

$$V = ?$$

CALCULATIONS.

$$V = \frac{kq}{r}$$

$$V = \frac{9 \times 10^9 \times 15 \times 10^{-9}}{16 \times 10^{-2}} = 843.7 \text{ Volt}$$

P.18 Find (a) the charge (b) the charge density on the surface of a conducting sphere of radius 15.2 cm and potential 215 V?

Sol.

DATA

$$V = 215 \text{ Volt}$$

$$r = 15.2 \times 10^{-2} \text{ m}$$

$$q = ?$$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{215 \times 15.2 \times 10^{-2}}{9 \times 10^9}$$

$$q = 3.6 \times 10^{-9} \text{ C}$$

$$(b) \quad \sigma = \frac{q}{A} = \frac{q}{4\pi r^2}$$

$$\sigma = \frac{3.6 \times 10^{-9}}{4\pi \times (15.2 \times 10^{-2})^2} = 1.25 \times 10^{-8} \text{ C/m}^2$$

P.19 Potential at the center of a uniformly charged circular disk of radius 3.5 cm is 550 V. What is the total charge in the disc.

Sol.

DATA

$$r = 3.5 \text{ cm} = 3.5 \times 10^{-2} \text{ m}$$

$$V = 550 \text{ V}$$

$$q = ?$$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{Vr}{k} = \frac{550 \times 3.5 \times 10^{-2}}{9 \times 10^9}$$

$$q = 2 \times 10^{-9} \text{ C}$$

